10 Measurement

We measure different things in our daily lives. For example, we measure the length, mass, and quantity of objects for different purposes. We also measure time using a clock and temperature of our body using a thermometer.

Learn about

- Measurement and its need
- Measurement of length
- Measurement of mass
- Measurement of capacity
- Measurement of time
- Measurement of temperature

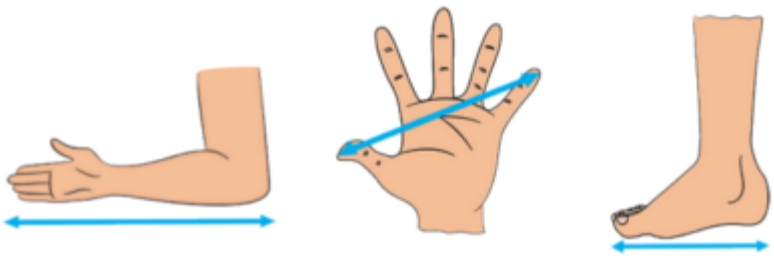
▶ Measurement and Its Need

Measurement *is defined as the process of finding out the size or the quantity of something.* We use different units to measure things. *A* ***unit*** *is defined as a fixed quantity used as a standard of measurement.*

What is the Need of Standard Units?

In earlier times, people used to measure the length of an object using handspan, cubit, or foot span.

For example, Nita measured the length of her study table and found that it is 18 handspans long. Her mother also measured it and found it to be lesser. The measurement varied as the size of palms of Nita and her mother were different.



Using body parts for measurement

These measures did not give correct measurements because different people have different measures of their body parts. Thus, they felt a need to have a fixed unit for measuring different things. These fixed units are called **standard units**.

Examples of measurement in our day-to-day life

- We can measure the length of an object, say a table, using a measuring tape.
- A fruit vendor measures the weight of fruits with the help of a weighing balance.
- We can measure time using a wall clock or a wristwatch.
- We can measure the quantity of medicine using the medicine bottle cup.
- A doctor measures the body temperature of an ill person using a thermometer.

Thus, measurements are a part of our daily routine. *The objects that are used for measuring different things are called* ***measuring instruments****.*

▶ Measurement of Length

Length *is the size or measure of how long an object is.*

The standard units used for measuring length are **metre** (m) and **centimetre** (cm). A metre ruler, metre scale, or measuring tape are some of the instruments that are used to measure length.

Measuring tape and a metre scale

Small lengths are measured in centimetres. Long lengths are measured in metres. Longer lengths or distances are measured in kilometres (km).

Think and Discuss

We cannot measure the distance between the Sun and the Earth in metres or kilometres. Discuss and find how is this distance measured.

Using a ruler and a measuring tape We use a ruler to measure the length of small objects, such as a pencil, pencil box, or a tiffin box. For example, to measure the length of a pencil, the ruler is kept along the pencil. One end of the pencil is kept at the 0 mark on the ruler.

Measuring the length of a pencil

Suppose the other end of the pencil reaches the 9 cm mark, then the length of the pencil is 9 cm. Likewise, if the pencil tip reaches up to 9 cm and 3 small divisions more on the ruler, then the length of the pencil will be 9.3 cm (9 cm and 3 mm).

We should keep our eyes exactly above the ruler and check the marking on the ruler that coincides with the other end of the pencil otherwise we will get incorrect measurement.

In a similar manner, a measuring tape is used to measure the length of bigger objects such as the length of a ribbon, rope, or cloth.

Questions

APPLYING

Choose the correct unit to measure the following objects.

1. Length of your notebook .....
2. Distance between two cities .....
3. Length of your class .....
4. Length of your eraser .....

Measurement of Mass

Mass *is defined as the amount of material an object contains.*

Kilogram (kg) and **gram** (g) are the standard units for measuring mass. Small masses are measured in grams while large masses are measured in kilograms. The mass of an object is measured using a **weighing machine** or a **weighing balance**.

Using a Weighing Machine

TP

We can measure the mass of our body using a weighing machine. The stepwise method to find our mass is as follows:

- First, ensure that the needle of the weighing scale coincides with the zero mark.
- Now, stand on the weighing machine barefoot. This will change the reading on the dial.
- Take the help of an adult to read the fixed position of the needle.

Weighing machine

This reading will be your mass in kilograms.

Using a Weighing Balance

A weighing balance measures the mass of an object by comparing it with the mass of a known weight. For example, while selling fruits and vegetables, the vendor places the

Weighing balances

vegetables or fruits on one pan and puts the known weights on the other pan of the weighing balance. The mass of both the objects will not be same until the needle of the weighing balance comes at the centre.

They can also use beam balance for weighing. In the beam balance, the plates are hung vertically with ropes attached to the middle rod.

▶ Measurement of Capacity

*The **capacity** of a container is defined as the amount of liquid it can hold.* The standard units for measuring capacity are **litre** (L) and **millilitre** (mL). Measuring cylinders and beakers is used to measure capacity.

Beakers and measuring cylinder containing solutions

▶ Measurement of Time

Time is the measure of how long does an action takes. It is measured in **hours**, **minutes**, and **seconds**. Their symbols are **h**, **min**, and **s**, respectively.

Using a Clock

A clock helps us to read time. Most of the clocks have three hands—**hour**, **minute**, and **second** hand.

Among the three hands, the hour hand moves the slowest and indicates the time in hours. The minute hand shows the time in minutes and moves faster than the hour hand. The second hand moves the fastest and shows the time in seconds.

Wall clock and digital alarm clock

Nowadays, digital clock are also in use. These clocks help in reading the time quickly and easily.

▶ Measurement of Temperature

***Temperature** is a measure of how cold or hot an object is.* Temperature is measured using a thermometer. It is usually measured in **degree Celsius** (°C) or **degree Fahrenheit** (°F).

Clinical thermometer

Using a Thermometer

A thermometer is used to measure temperature. The thermometer used for measuring body temperature is a **clinical thermometer**. The stepwise method to measure the temperature is as follows:

- Always clean the thermometer before use.
- Shake it well such that the mercury inside the thermometer moves into the bulb.
- Place the bulb of the thermometer in the armpit for one to two minutes. This will cause mercury to move along the scale printed on the thermometer and become stable. The stable reading is your body temperature.

  The normal body temperature is 98.6 °F or 37 °C.

Think and Discuss

Mercury thermometers are being replaced by digital thermometers. Think and discuss the reason for the same.

Case Study

Ramu, Pranav, and Mohit were admitted to Vivekanda Hospital on the same day due to the recent heat wave in Delhi. Before giving them any treatment, the nurse checked their body temperatures were 101 °F, 98.6 °F, and 103.4 °F, respectively.

Based on the above information, answer the following questions:

a. Name the instrument used to measure the body temperature.

b. Name the patient who had a normal body temperature.

c. Name the patient who had a high fever.

Questions

Write the standard unit to measure the following.

1. Mass .....
2. Capacity .....
3. Length .....
4. Temperature .....
5. Time .....

Wrap Up

- Length is the size or the measure of how long an object is. The units for measuring length are kilometre (km), metre (m), and centimetre (cm).
- Mass is the amount of material an object contains. It can be measured in kilogram (kg) and gram (g).
- Capacity is the amount of liquid a container can hold. Litre (L) and millilitre (mL) are some units to measure capacity.
- Time is the measure of how long does an action takes. The units for time are hours (h), minutes (min), and seconds (s).
- Temperature is a measure of how hot or cold an object is. It can be measured in degree Celsius (°C) or degree Fahrenheit (°F).

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A **Choose the correct option.**

1. Fixed quantity used as standard of measurement is called a/an ..... .
   a. handspan b. foot span c. cubit d. unit
2. The standard unit of length is ..... .
   a. metre b. gram c. litre d. centigrade
3. Amount of material an object contains ..... .
   a. length b. mass c. capacity d. temperature
4. The measuring cylinder measures ..... .
   a. capacity b. length c. mass d. temperature
5. The thermometer measures ..... .
   a. length b. mass
   c. temperature d. capacity

B **Assertion and Reasoning questions.**

1. **Assertion (A):** The actual length of an object is not measured correctly if it is not kept along the zero mark of the ruler.

Reason (R): There are no markings before the zero mark on the ruler.

a. Both A and R are True
b. Both A and R are False
c. A is True and R is False
d. A is False and R is True

2. **Assertion (A):** We can measure the amount of water we drink using a beaker.

   Reason (R): Beakers are used to measure capacity.

   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True

C Write *T* for True and *F* for False. Correct the False statements.

1. Second is the standard unit for measuring mass.
2. Handspan was used in the past to measure length.
3. Measuring tape is used to measure the length of a rope.
4. A measuring cylinder is used to measure capacity.
5. We can check our height on a weighing scale.

D Match the following.

Column A	Column B
1. Thermometer	a. Time
2. Weighing machine	b. Amount of water
3. Watch	c. Temperature
4. Measuring tape	d. Mass of a sack of flour
5. Measuring cylinder	e. Length of a table

E Name the following.

1. Instrument used to measure body temperature .....
2. The process of finding out the size or quantity of an object .....
3. Instrument used to measure capacity .....
4. A clock is used to measure it .....

F Write the symbols for the following units.

1. Gram .....
2. Degree Celsius .....
3. Metre .....
4. Kilogram .....
5. Litre .....

6. Millilitre .....
7. Centimetre .....

SECTION II

G Short answer questions.

1. Define a unit.
2. What is length?
3. When we buy potatoes, what is the standard unit used to weigh potatoes?
4. What is temperature? Which instrument is used to measure temperature?

H Long answer questions.

1. Give four examples of measurement from our day-to-day life.
2. Will the length of your study table be the same if you or your parents measure it using your respective footspan? Based on this, explain why do we need standard units for measuring things.
3. Suppose you bought coloured pens for your new assignment. But the pens do not fit into your stationery box. Write the steps you will follow to measure the length of the coloured pens and the stationery box.
4. How does a vegetable vendor weigh the vegetables?
5. Define capacity. Name any two standard units to measure capacity.

Picture Study

Look at the picture and answer the following questions.

1. Name the instrument shown.
   .....
2. What does it measure?
   .....
3. What are the standard units used to measure it?
   .....
4. Write the steps for measuring it.
   Step 1: .....
   Step 2: .....

My Learning Corner

A Think about

1. Pratibha works in 'Triveni Cloth House'. Name the instrument which she uses to measure length of the clothes, and which standard unit does she use while measuring it.

B Try out

1. You may have learnt in Mathematics that three-dimensional shapes like cube, cuboid, etc., have three dimensions – length, breadth, and height. Divide the class into groups and conduct activities to measure the dimensions of some 3D objects in your class such as table, chair, etc. Write the name of five more 3D objects and note down their measurements.
2. Project Work: (Do it in your Science Scrapbook.)

   Visit a nearby market with your mother, with a notepad and a pencil. Make a list of different kinds of shops that you visit. Note down the names of any five shops, the things your mother purchases from each shop, and the measuring instruments used by the shopkeeper. Also, write down the names of the standard units used while purchasing. Make a table as given below.

	Name of shop	Things purchased	Measuring instruments	Standard unit of measurement
1				
2				

3. In the classroom, with the help of your class teacher, conduct a 'Science Quiz for Measurement' using flash cards.

   Steps to conduct the quiz:
   - Show flash cards of different measuring instruments and ask their names.
   - Show flash cards with 'Length/Mass/Capacity/Time/Temperature' written on them and ask students which measuring instrument measures each of them.
   - Show flash cards of measuring instruments and then ask which standard units are used when measuring with the help of these instruments.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• explain the need of measurement.	
• explain measurement of length and how do we measure length.	
• explain measurement of mass and how do we measure mass.	
• explain measurement of capacity and how do we measure capacity.	
• explain measurement of time and how do we measure time.	
• explain measurement of temperature and how do we measure temperature.	
• draw pictures of different measuring instruments.	

Worksheet

Pictures of some measuring instruments are shown below. Write their names and the measuring units used while measuring with these instruments. Also, draw and colour them in the space provided.

	Measuring instruments	Draw and colour here.	Name of instrument and standard measuring unit
1.			....
2.			....
3.			....
4.			....